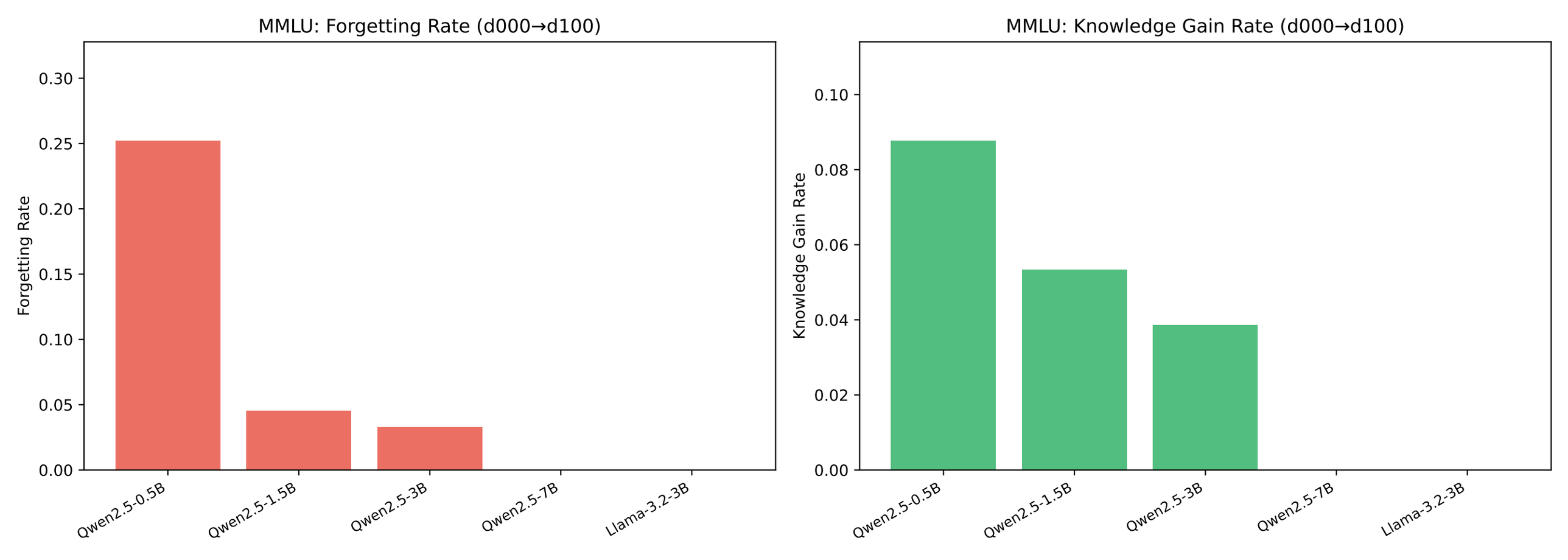
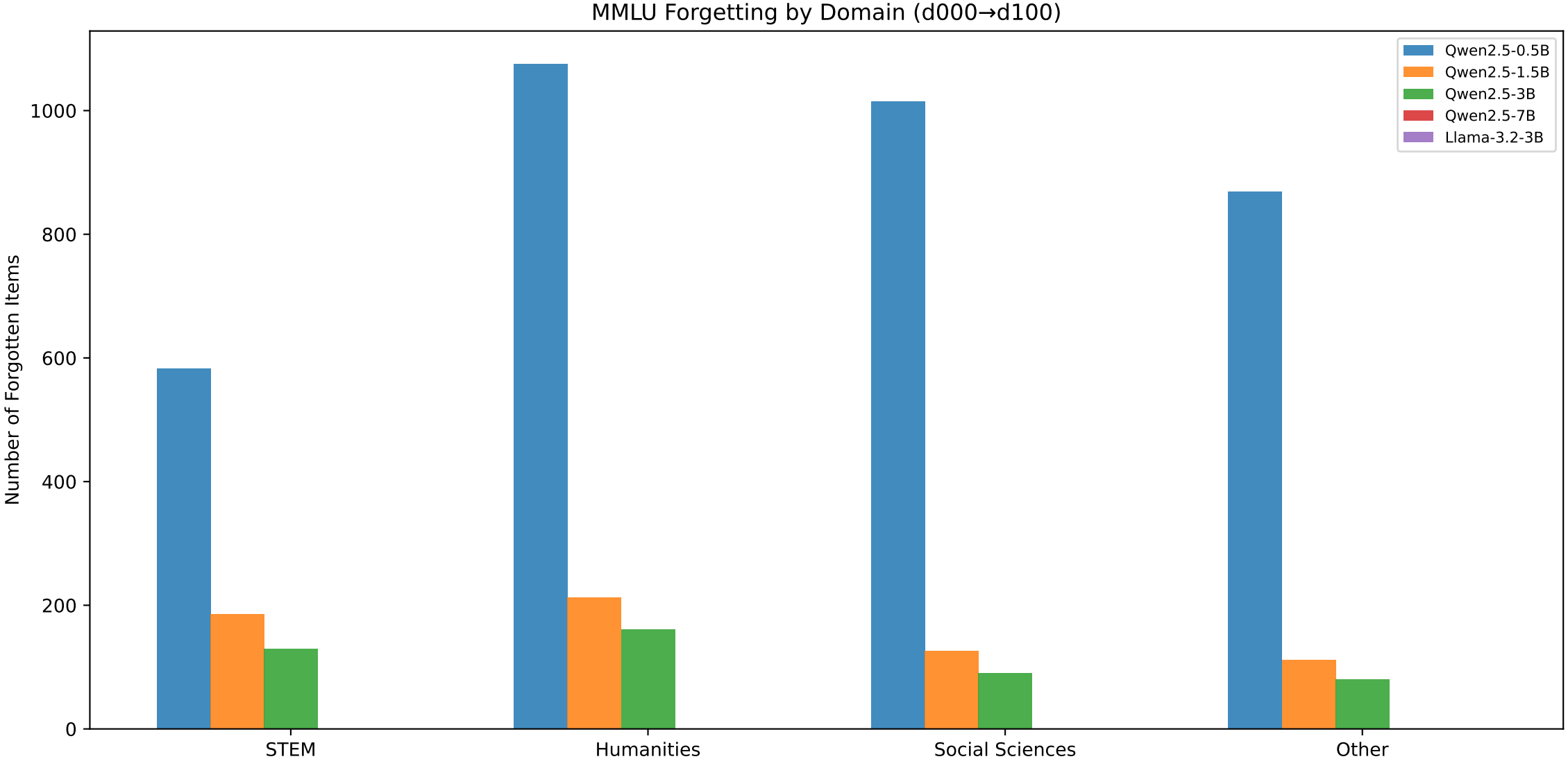
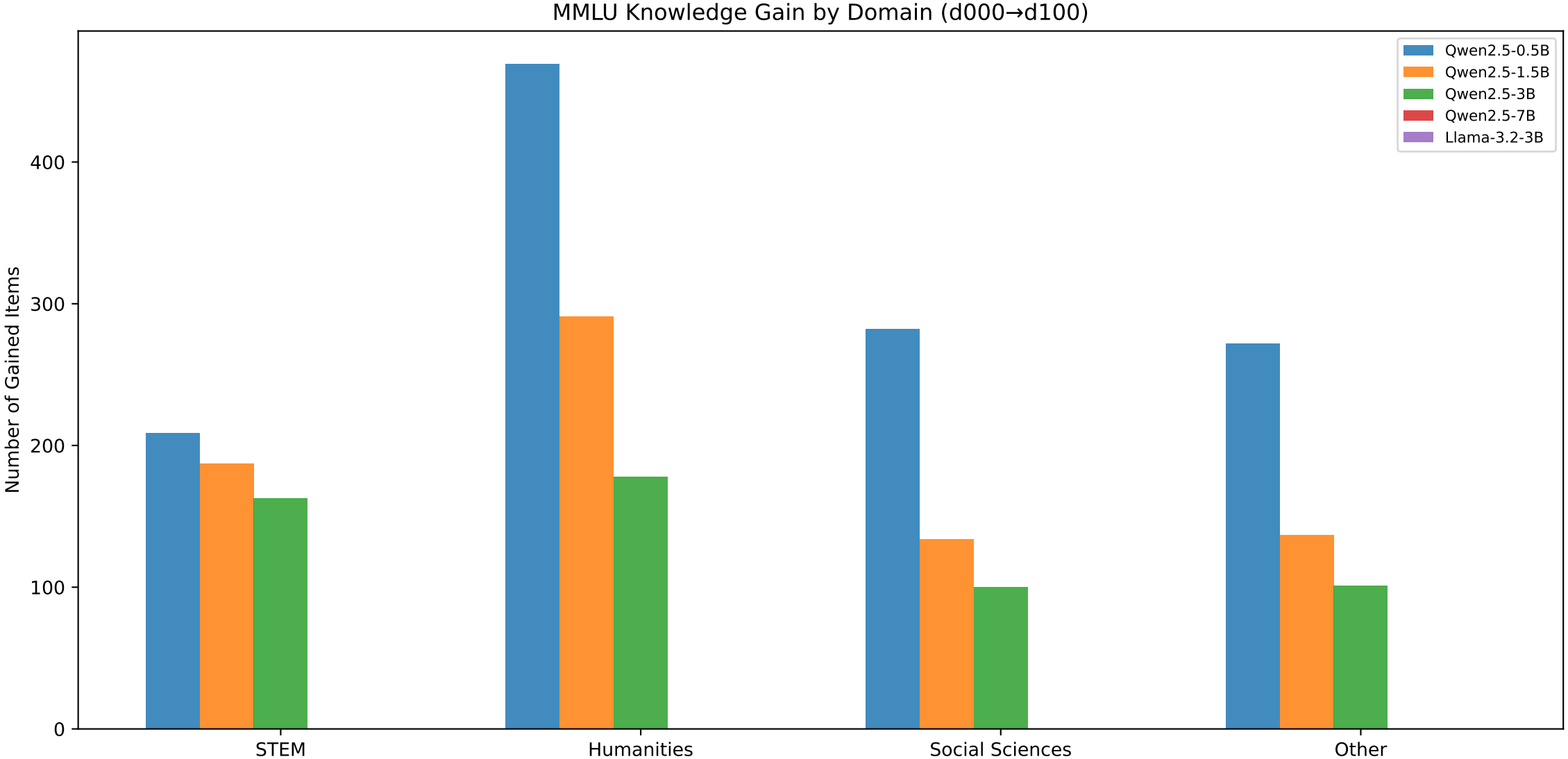
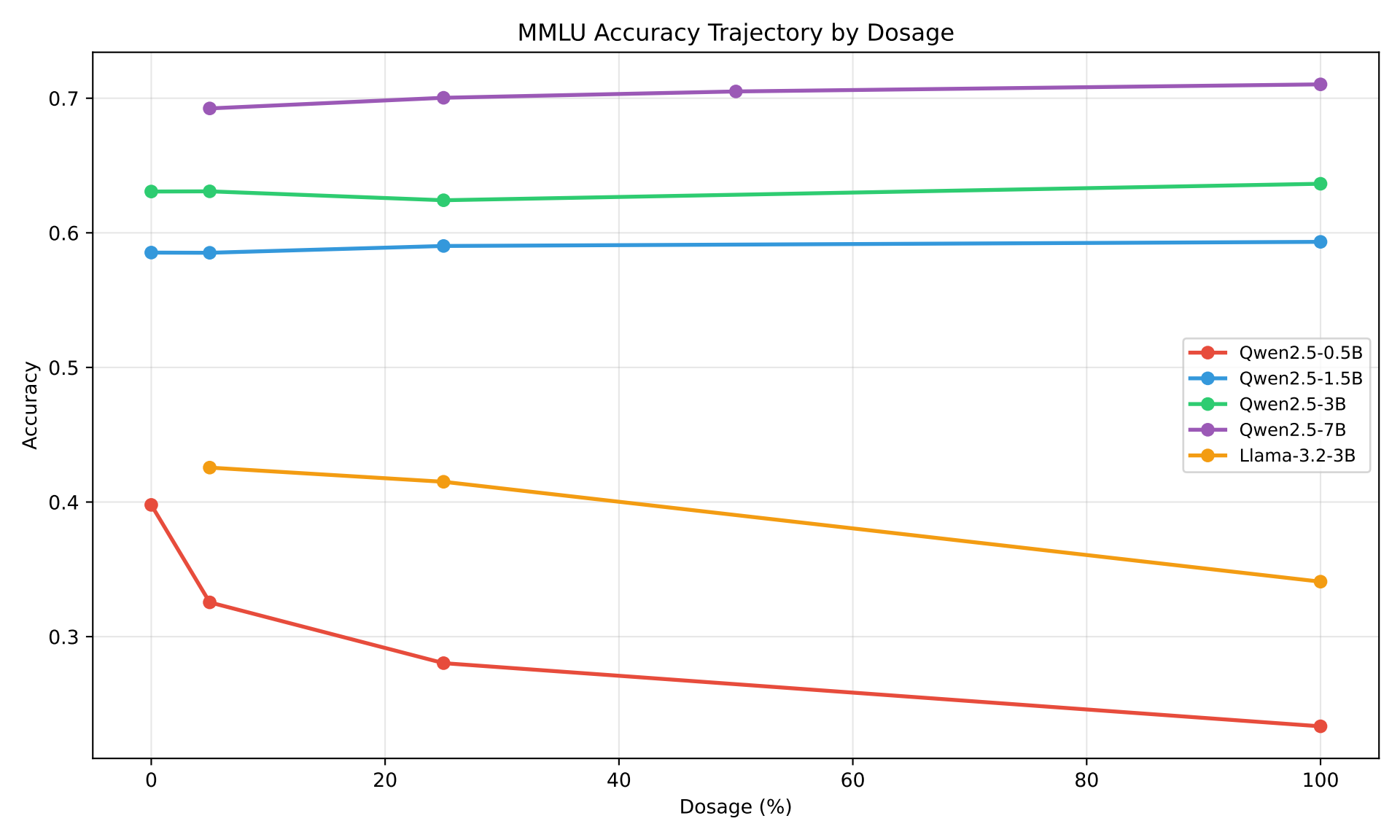
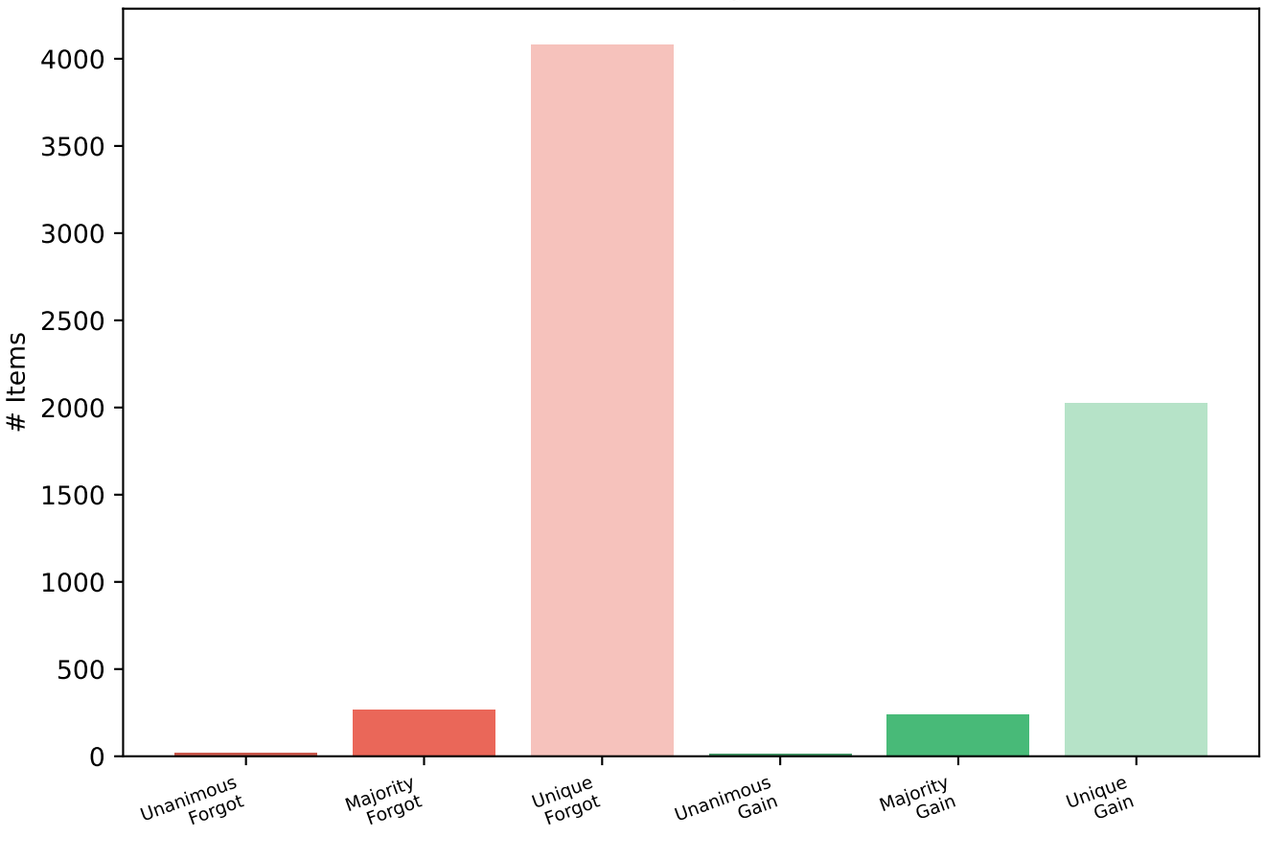
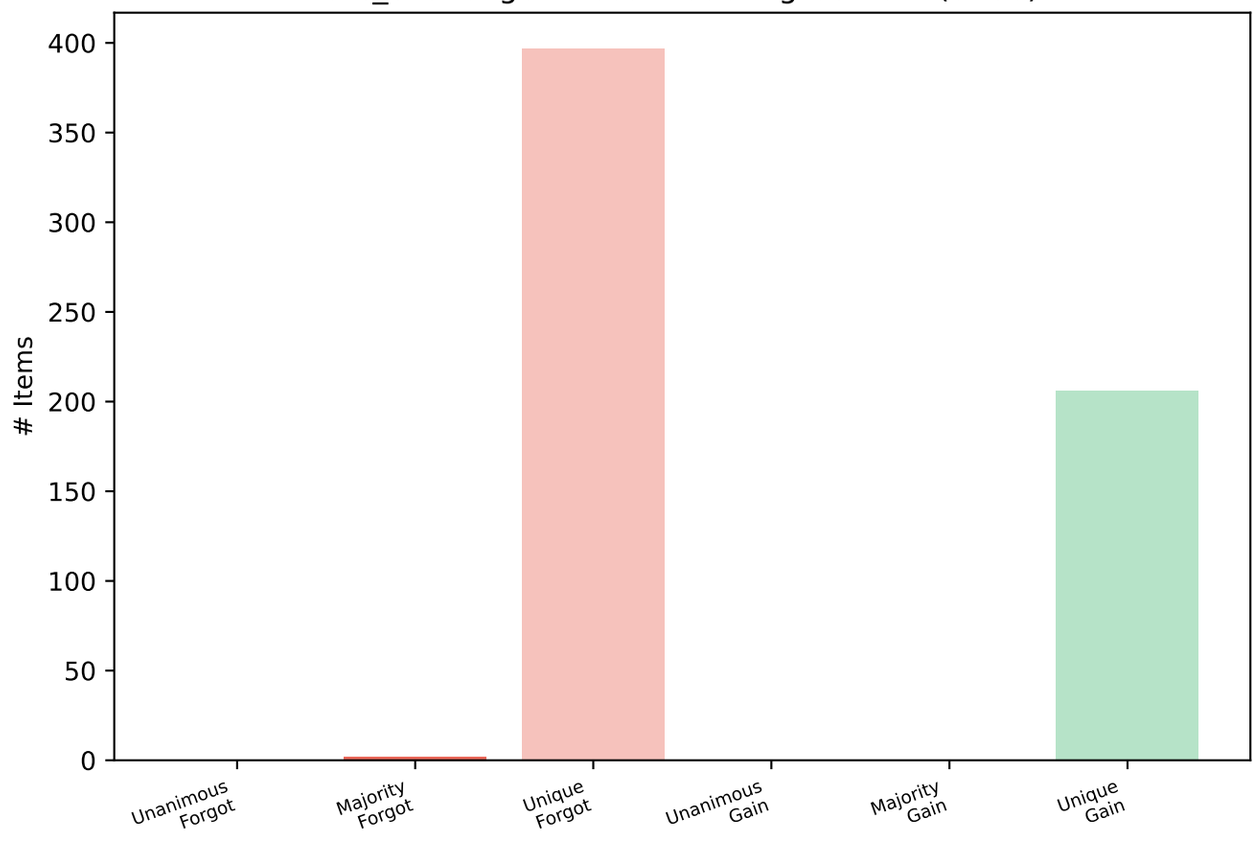

MMLU: Forgetting Rate (d000→d100)
Forgetting Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-3B
MMLU: Knowledge Gain Rate (d000→d100)
Knowledge Gain Rate
0.00
0.02
0.04
0.06
0.08
0.10
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-3B

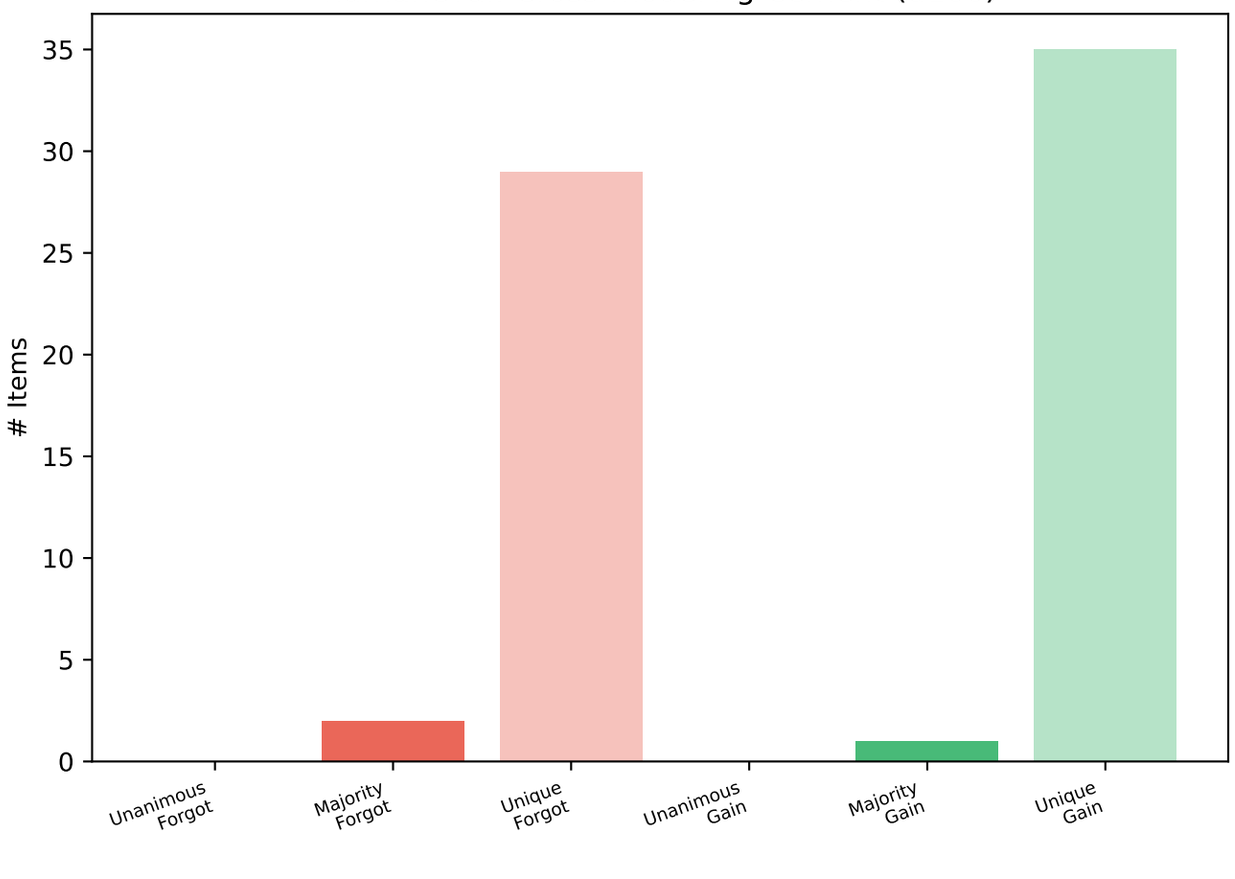

MMLU Forgetting by Domain (d000→d100)
Number of Forgotten Items
0
200
400
600
800
1000
STEM
Humanities
Social Sciences
Other
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-3B

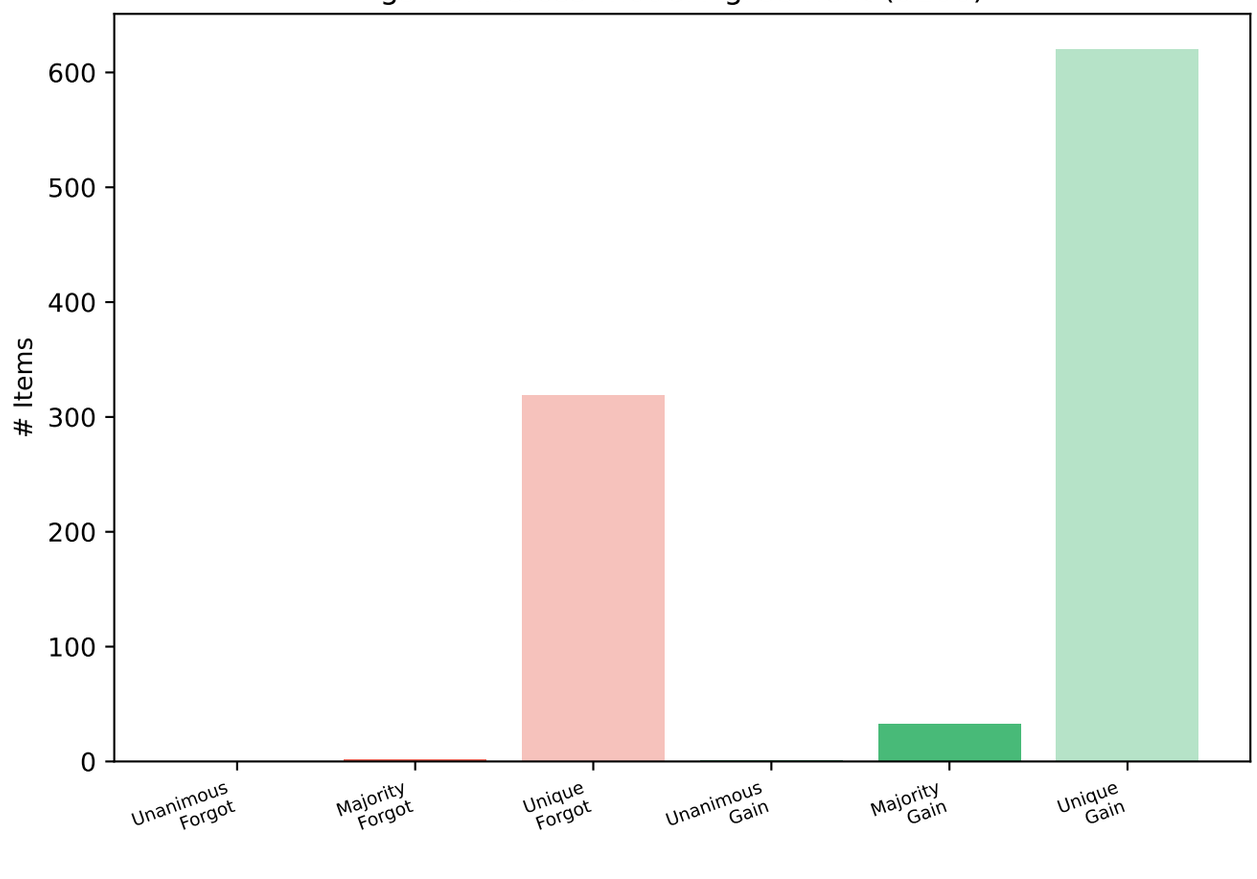

MMLU Knowledge Gain by Domain (d000→d100)
Number of Gained Items
0
100
200
300
400
STEM
Humanities
Social Sciences
Other
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-3B

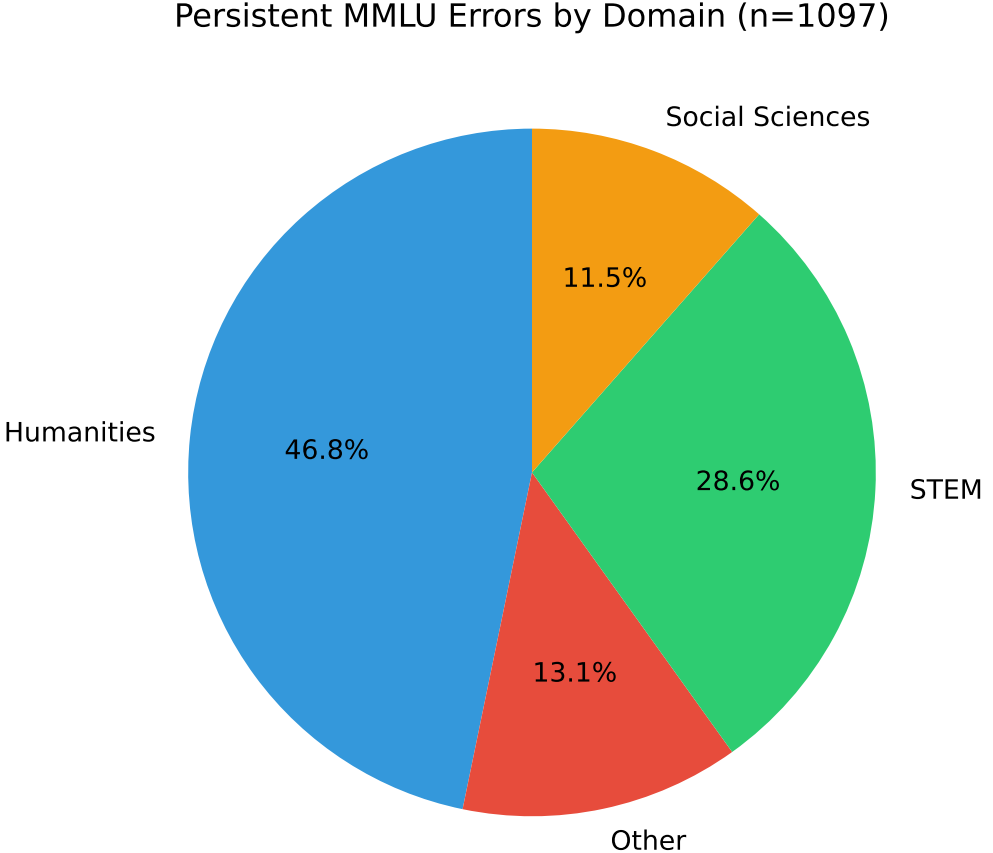

MMLU Accuracy Trajectory by Dosage
Accuracy
Dosage (%)
0.7
0.6
0.5
0.4
0.3
0
20
40
60
80
100
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-3B

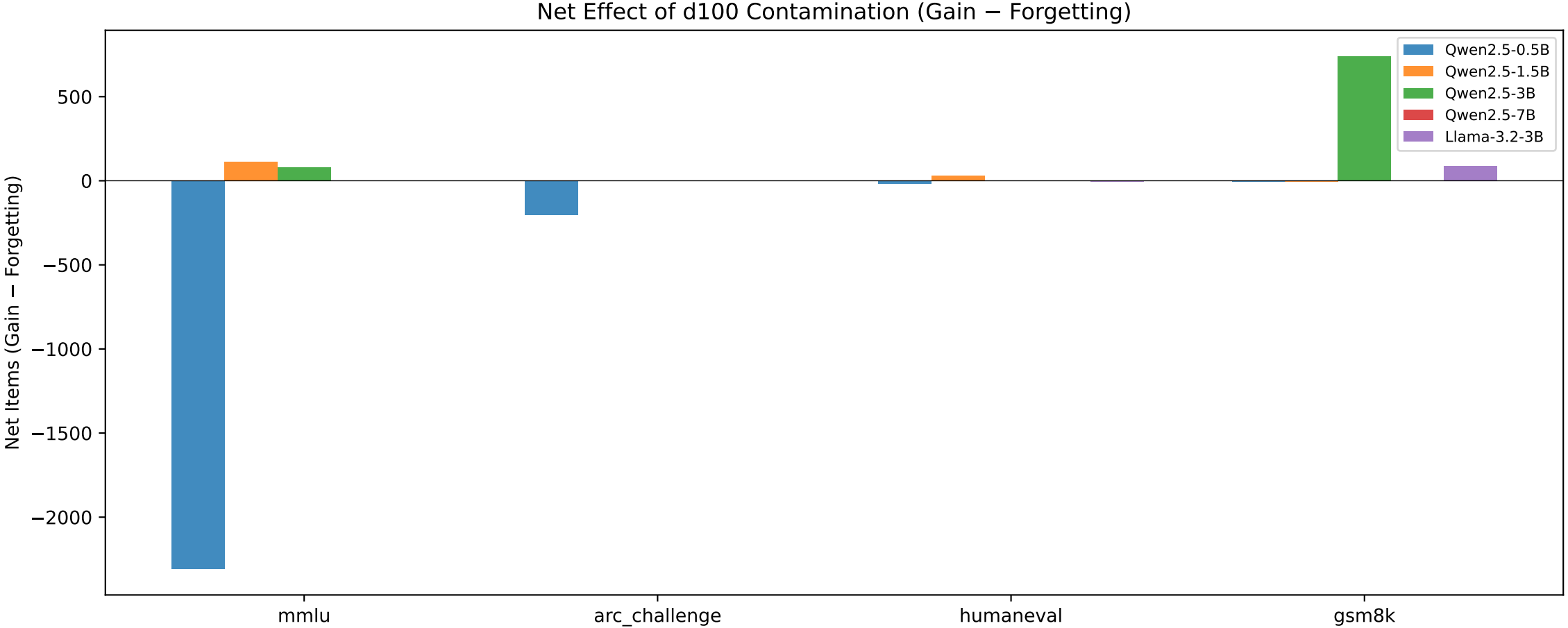

mmlu: Cross-model Agreement (d100)
Items
0
500
1000
1500
2000
2500
3000
3500
4000
Unanimous Forgot
Majority Forgot
Unique Forgot
Unanimous Gain
Majority Gain
Unique Gain


arc_challenge: Cross-model Agreement (d100)
Items
0
50
100
150
200
250
300
350
400
Unanimous Forgot
Majority Forgot
Unique Forgot
Unanimous Gain
Majority Gain
Unique Gain


humaneval: Cross-model Agreement (d100)
Items
0
5
10
15
20
25
30
35
Unanimous Forgot
Majority Forgot
Unique Forgot
Unanimous Gain
Majority Gain
Unique Gain


gsm8k: Cross-model Agreement (d100)
Items
0
100
200
300
400
500
600
Unanimous Forgot
Majority Forgot
Unique Forgot
Unanimous Gain
Majority Gain
Unique Gain


Persistent MMLU Errors by Domain (n=1097)
Social Sciences
11.5%
Humanities
46.8%
28.6%
STEM
13.1%
Other


Net Effect of d100 Contamination (Gain − Forgetting)
Net Items (Gain − Forgetting)
500
0
−500
−1000
−1500
−2000
mmlu
arc_challenge
humaneval
gsm8k
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-3B